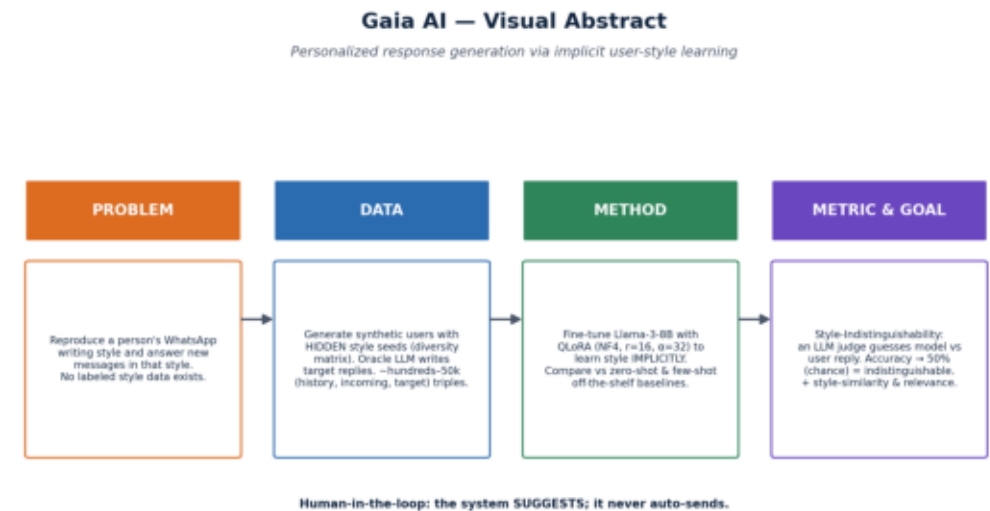


Gaia AI — Interim Report

Dataset, EDA & baseline · Adir · LLM course (HIT)

1 · Project Review

- Goal: generate WhatsApp replies in a user's own implicit style, from their chat history.
- Changed from proposal: added a MULTILINGUAL style-similarity embedder (data is he+en); hardened synthetic generation.
- Novelty/contribution:
 - Implicit style learning (no tone labels) from synthetic personas with hidden seeds.
 - Evaluation by Style-Indistinguishability (LLM judge → 50%) rather than a soft similarity score.



2 · Previous Work

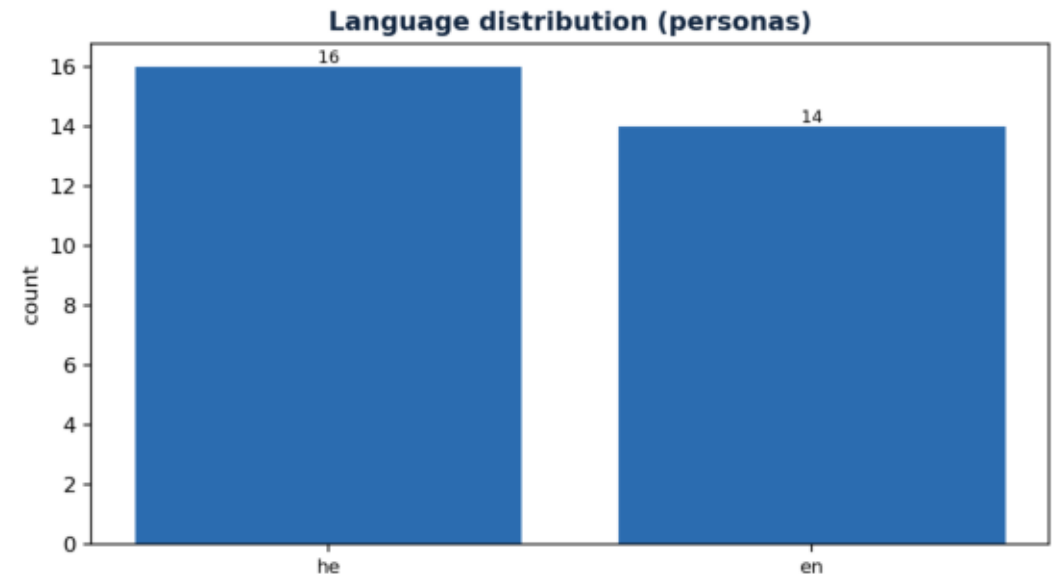
- Reviewed across our four pillars (full table + citations in docs/related_work.md):

Paper (year)	Task	Relation to us
QLoRA (2023)	4-bit LoRA fine-tuning	Our exact training method
Self-Instruct (2023)	LLM-generated training data	Our synthetic-data strategy
LLM-as-a-Judge (2023)	LLM eval + biases	Basis of our indistinguishability metric
LaMP (2024)	Personalization benchmark	We learn style implicitly, not via retrieval
PersonaChat (2018)	Persona-grounded dialogue	We use implicit, not explicit, personas

Recent, high-citation works (NeurIPS/ACL). arXiv IDs in docs/related_work.md.

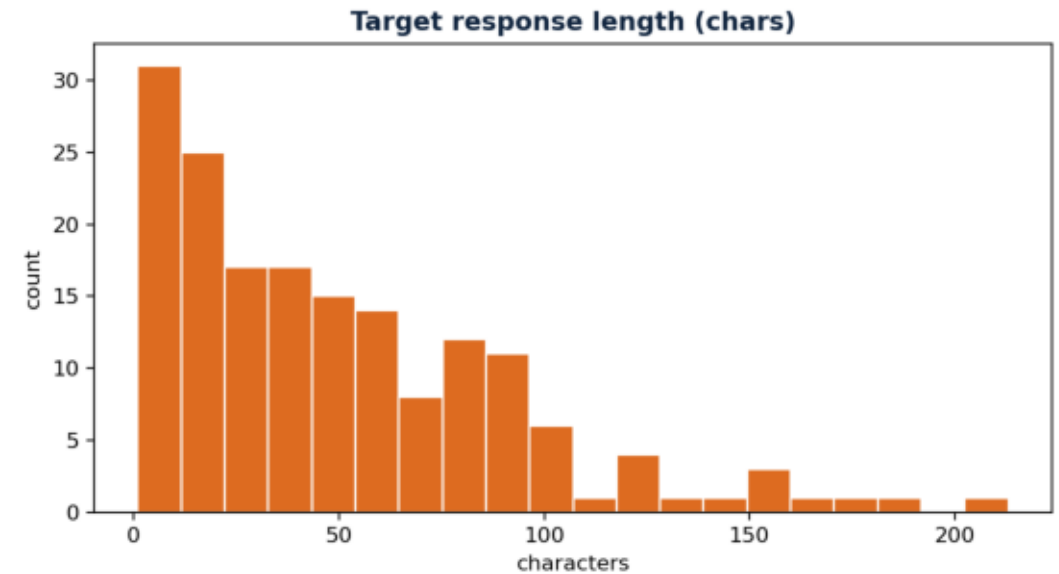
3 · Dataset & EDA

- Synthetic dataset generated with an LLM (Self-Instruct-style), bilingual he/en.
- Each user has a HIDDEN style seed from a 10-axis diversity matrix; targets written by an oracle that sees the seed.
- Splits are PER-USER (no user in both train and test) → tests generalization to unseen people.
 - Scale: 30 personas · 22 histories · 170 (history, incoming, target) pairs.
 - Target reply length: mean 50.2 chars (median 41.0, range 1-213).
 - History message length: mean 39.5 chars over 461 messages.



3b · EDA — length & style distributions

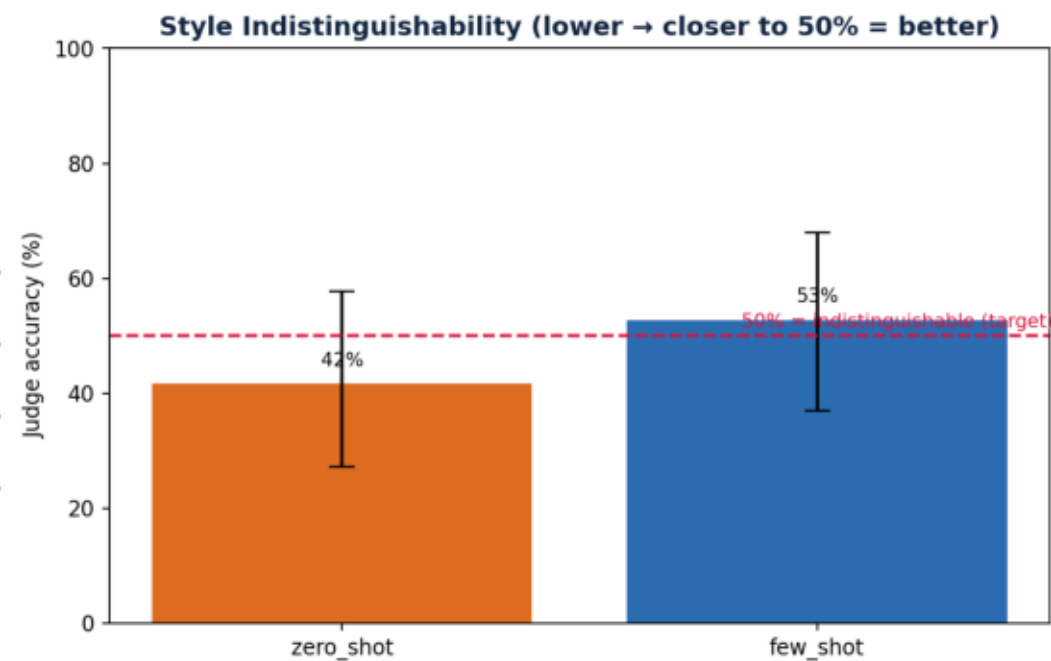
- Style axes are spread across the diversity matrix (formality, emoji, slang, length).
- Message and target-reply lengths skew short, matching real WhatsApp behavior.



4 · Baseline Solution & Results

- Baselines (off-the-shelf, no fine-tune): zero-shot and few-shot prompting on Llama-3.x via Groq.
- Oracle = upper bound (LLM that saw the hidden seed).
- Evaluated on the held-out per-user test split with all three metrics.

Model	Indist.	95% CI
zero_shot	41.7%	[27,58]%
few_shot	52.8%	[37,68]%
fine_tuned	pending (GPU)	—



Real baseline numbers (zero/few-shot via Groq). fine_tuned = run on GPU.

5 · Plan

- Remaining steps to the final report:

Step	Scope	Expected outcome
Fine-tune	QLoRA Llama-3-8B on cloud GPU (Modal/RunPod)	Trained adapter
Re-eval	Add fine_tuned column vs baselines	Full comparison table
Scale data	More users (paid Groq tier / multi-day)	More robust numbers
Error analysis	Inspect bad cases, tune prompts	Quality improvements
Final deck	Results + conclusions	Submission